

Subject Description Form

Subject Code	ITC6862RT																																																				
Subject Title	Guided Study in Polymer Science																																																				
Credit Value	3																																																				
Level	6																																																				
Pre-requisite / Co-requisite / Exclusion	Nil																																																				
Objectives	To introduce the concepts and fundamentals of polymer science to students who have no prior knowledge in the area																																																				
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. know the history of polymers b. recognize the structure-property relationship c. comprehend the importance of molecular weights d. comprehend chain-reaction and step-reaction polymerization e. comprehend copolymerization f. relate polymer science to textiles																																																				
Subject Synopsis/ Indicative Syllabus	a. Introduction to polymer science – history, technical terms b. Polymer structure-property relationship c. Rheology and solubility d. Molecular weight e. Testing and characterization f. Step-growth polymerization g. Chain-growth polymerization h. Copolymerization i. Applications in textiles																																																				
Teaching/Learning Methodology	The students are required to read selected book chapters and papers accompanied with guided questions. The importance of each topic will be discussed. Relevant videos and demonstration will be used to help elaborate the subject.																																																				
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>1. 6 assignments</td><td>30%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Presentation</td><td>15%</td><td></td><td></td><td></td><td></td><td></td><td>✓</td></tr><tr><td>3. Examination</td><td>55%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table> The continuous assessment includes an assignment on each topic and a presentation. A written examination will be conducted at the end of the semester.							Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed						a	b	c	d	e	f	1. 6 assignments	30%	✓	✓	✓	✓	✓	✓	2. Presentation	15%						✓	3. Examination	55%	✓	✓	✓	✓	✓	✓	Total	100%						
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2. Presentation	15%						✓																																														
3. Examination	55%	✓	✓	✓	✓	✓	✓																																														
Total	100%																																																				

Student Study Effort Expected	Class contact:	
	▪ Discussion	17 Hrs.
	▪ Videos and demonstrations	9 Hrs.
	Other student study effort:	
	▪ Reading	74 Hrs.
	▪ Assignments	20 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	1. Billmeyer, W. Textbook of Polymer Science (3 rd edition) Wiley (1984) 2. Brydson, J.A. Plastics Materials (7 th edition) Elsevier (1999) 3. Campbell, I.M. Introduction to Synthetic Polymers, Oxford Science Publications (1994) 4. Challa, G. Polymer Chemistry, Wiley (1993) 5. Cowie, J.M.G., Arrighi, V. Polymers: Chemistry and Physics of Modern Materials (3 rd edition) CRC Press (2007)	